ISSUE 01 · POLAR / ICE CITY · A CITY BUILT IN 3D

AURORA HAVEN

A polar city that thrives



The city's own sun

A compact fusion station keeps 45,000 people warm through the polar night

PAGE 14

A warm heart inside the ice

Four geothermal spa levels hidden in a 35-metre snow peak

PAGE 19

Bread and water from snow and air

Inside the plant that feeds a city with no farmland

PAGE 17

−45 °C. 120 km/h. No sun.

The polar challenge, and the five ideas that answer it

PAGE 6



INSIDE THIS ISSUE

Contents

Shops and cafés on a downtown street at dusk. The streets are warmed from below, so they stay clear of ice without salt.

4 **Welcome to Aurora Haven**
A compact city of 45,000 under a snow peak

6 **The Polar Challenge**
Cold, darkness, wind and permafrost

7 **Five Design Principles**
The ideas behind every building

8 **The Whole City Plan**
From the wind farm to Mount Aurora

10 **Nine Blocks, One Short Walk**
Downtown around the Aurora Roundabout

13 **A Livable City Every Day**
Homes, schools, health, shops and culture

14 **The City's Own Sun**
Fusion Power Station

17 **Bread and Water from Snow and Air**
Snow-to-Food & Water Plant

19 **A Warm Heart Inside the Ice**
Mount Aurora Geothermal Spa Resort

23 **Three Districts, One Loop**
How heat, water, oxygen and waste go round

24 **Energy for the Polar Night**
Fusion, wind, sun, geothermal and storage

25 **Nothing Is Wasted**
Water and waste in closed loops

26 **Getting Around Fossil-Free**
Metro, trams, heated roads and the airport

27 **Salad Under the Northern Lights**
Inside the Evergreen Greenhouse Dome

28 **Safe and Ready for Any Storm**
Rescue, shelters and blizzard forecasts

29 **Take the Tour**
Eight stops around the city

30 **Places to Visit**
35 key places and what they do

31 **Aurora Haven by the Numbers**
The key figures in one place

“Our goal was not only to survive the cold but to build a city where people really want to live.”



The Aurora Lagoon, a 38 °C open-air mineral pool at the foot of Mount Aurora, under the northern lights.

WELCOME TO THE FIRST ISSUE of Aurora Haven, a magazine about our city for the Polar / Ice environment.

Aurora Haven is a compact city of about 45,000 people on a frozen tundra plateau, below a 35-metre snow peak called Mount Aurora. Our goal was not only to survive the cold but to build a city where people really want to live.

In these pages you will find the challenge we faced, our design ideas, our three signature districts and everyday life in the city.

Every picture in this issue was rendered from a 3D model of the city. At the back, a self-guided tour takes you around the city, from the monument to the rescue station.

[Group name]

[Member names] · [Class] · [Date]



ON THE COVER

Aurora 81 – Polar Landmark Tower, named in honour of Landmark 81 in Ho Chi Minh City, beside the street that leads to the glowing terraces of the Mount Aurora spa, under the northern lights. Its nine bundled glass tubes step back like bamboo stalks and cut blizzard wind load by about 25%.



CITY OVERVIEW

Welcome to Aurora Haven

Designed not just to survive the polar night, but to make life here warm, healthy and sustainable.

AURORA HAVEN IS a compact polar city of around 45,000 people, built on a frozen tundra plateau beneath the snow peak of Mount Aurora. Winters fall to −45 °C, the sun disappears for two months of polar night, and almost nothing can grow.

The whole city is shaped around one idea: heat, energy, water and food should never be wasted. Every building shares a single heating network, the city runs on zero-carbon energy, and nearly all of its water and waste is recycled.

Homes, schools, healthcare, shops and parks sit within a short walk of each other, connected by a downtown metro, electric trams and heated streets.



Aurora Haven at dusk, seen from the south-east: the fusion station in front, downtown and Aurora 81 in the middle, the wind farm on the windy northern edge.

What makes Aurora Haven unique is its three signature districts. A fusion power station acts as the city's own sun, giving steady power and heat even through the polar night. A Snow-to-Food & Water Plant turns snow and air into clean drinking water and food, so the city can feed itself without farmland.

And inside Mount Aurora, a geothermal spa resort offers warmth, rest and a view of the northern lights. Together they form one closed loop that keeps the city powered, fed and happy in one of the harshest places on Earth.

45,000
residents

−45 °C
winter lows

35 m
Mount Aurora, the snow peak

10 min
walk to daily needs



THE PLACE

The Polar Challenge

One of the hardest places on Earth to build a city, so every building had to answer one question.

Aurora Haven sits in one of the hardest environments on Earth. In winter the temperature falls to minus 45 degrees. For about two months the sun never rises, and for two months in summer it never sets. Blizzards can reach 120 kilometres per hour.

The ground is permafrost, so a warm building can thaw it and crack its own foundations. There is almost no farmland, and in deep winter supplies can only arrive by air.

So the designers asked one question about every building in the city:

"How does it help people stay warm, fed, safe and happy here?"

–45 °C
winter temperatures

2 months
of polar night, and two of midnight sun

120 km/h
blizzards

Permafrost
frozen ground that heaves under heat

A blizzard sweeps past the snowbreak pines towards the Ice-Proof Wind Farm on the windy northern edge.



Downtown in the low sun, with Aurora 81, the food plant silos and Mount Aurora behind.

THE IDEAS

Five Design Principles

Heat, energy, water and food should never be wasted.

01

Heat is life

One district-heating loop, fed by the fusion station and the geothermal plant, warms every building. Triple-glazed buildings on insulated piles do not thaw the permafrost.

02

Zero carbon

No fossil fuel burns anywhere in the city. Power comes from fusion, wind, sun and geothermal heat.

03

Closed loops

About 95% of water is recycled and rubbish is reused, so water, food and waste go round instead of being thrown away.

04

Compact and car-light

Daily needs are within a 10-minute walk, and the few cars are electric.

05

Resilient and healthy

Shelters keep every resident safe for 30 days, while light, parks, culture and a spa carry people through the polar night.

THE MAP

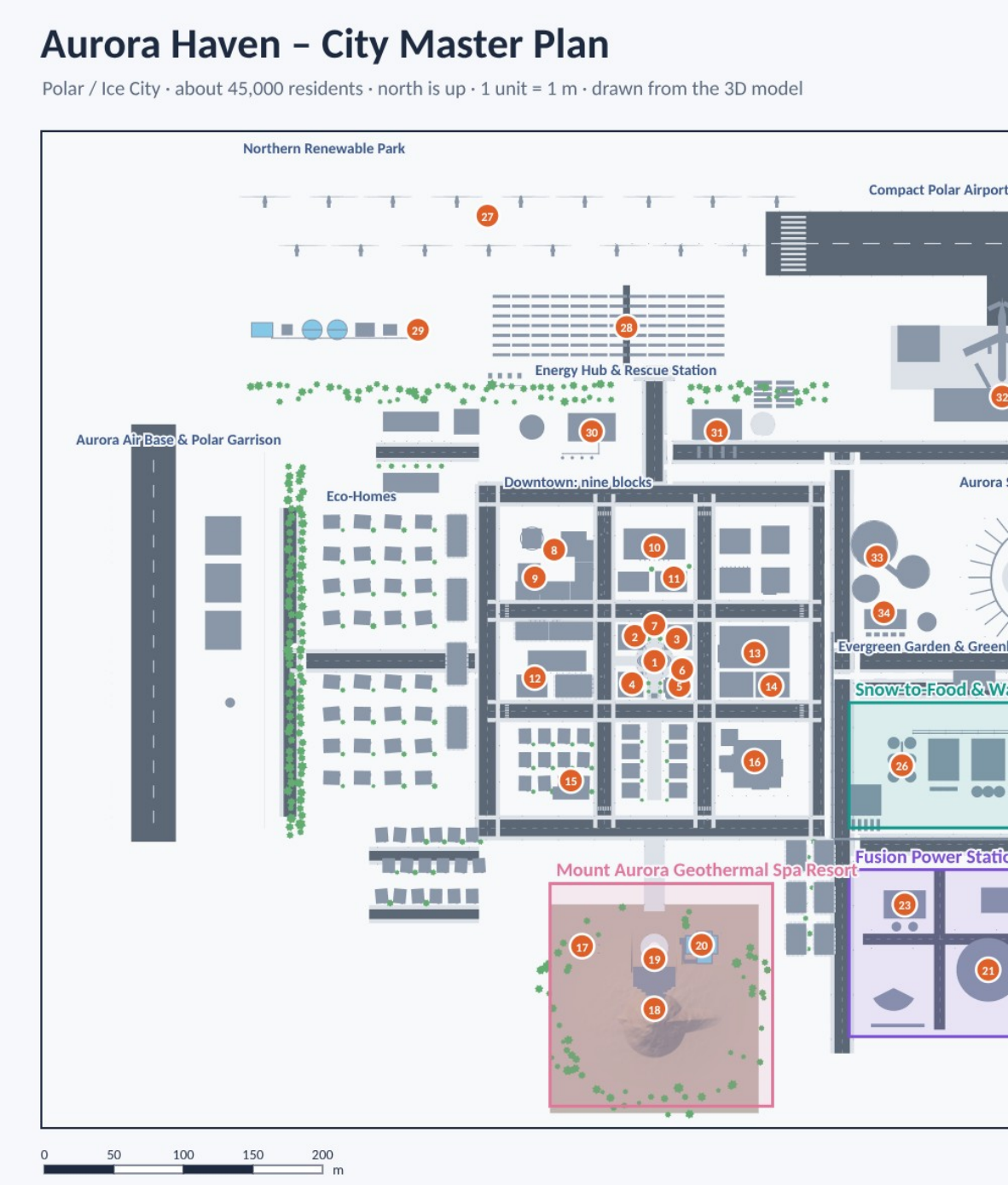
The Whole City Plan

DOWNTOWN

Nine blocks around the Aurora Roundabout and the President Ho Chi Minh Monument, with the hospital, schools, shops and the metro interchange.

NORTH

The windy north holds the Northern Renewable Park, the energy hub, the Polar Rescue and Resilience Station and the airport.



Aurora Haven – City Master Plan. North is up; one unit is one metre. Drawn from the 3D model; the numbers are the 35 places listed on page 30.

WEST

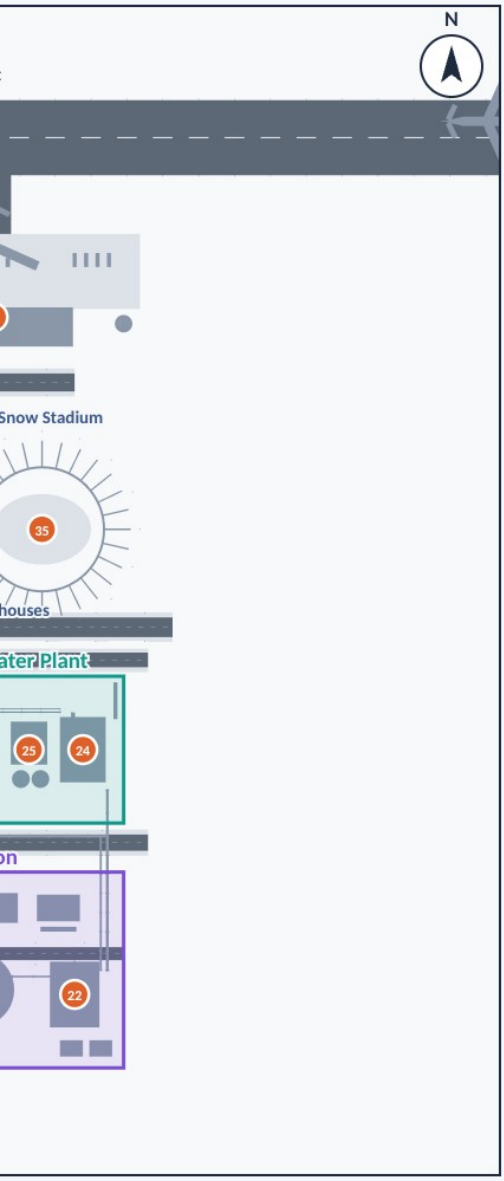
The air base and garrison, behind a belt of trees that screens the eco-homes.

EAST

Greenhouses, the snow stadium, the fusion station and the Snow-to-Food plant, linked by pipe bridges and heat mains.

SOUTH

Framed by Liberty Mall, Mount Aurora rises with its geothermal spa resort inside.



Signature districts

- Mount Aurora Geothermal Spa Resort
- Fusion Power Station
- Snow-to-Food & Water Plant

Map key

- Buildings and halls
- Roads and runways
- Plazas and paved yards
- Trees and parks
- Water and pools

Key places

1. President Ho Chi Minh Monument
2. Aurora Civic Center & Municipal Hall
3. Snowflake General Hospital
4. Northern Lights School & Botanical Conservatory
5. Igloo Fresh Supermarket & Food Hall
6. Aurora Cinema & Performing Arts Center
7. Underground Metro & Emergency Shelters
8. Aurora Core – AI Control Tower
9. Aurora Haven Police Headquarters
10. Arctic University – Main Hall
11. Aurora Public Library
12. West Market Fire & Ambulance Station
13. Aurora Glacier Mall & Crystal Atrium
14. Snow Lantern Primary School
15. Pinecone Kindergarten
16. Aurora 81 – Polar Landmark Tower
17. Aurora Snow Garden Park
18. Mount Aurora – Snow Peak & Hanging Glacier
19. Summit Sanctuary & Aurora Eye
20. Aurora Lagoon – Outdoor Geothermal Mineral Pool
21. Fusion Core – Compact D-T Tokamak
22. Power Conversion & Heat Export Hall
23. Plasma Waste Converter
24. Snow Harvest & Melt Hall
25. Drinking-Water & Mineral Works
26. Bioreactor Silos
27. Ice-Proof Wind Farm
28. Bifacial Solar Farm
29. Water Recycling Line – Recycled Water Tower
30. Geothermal Power Plant
31. Polar Rescue and Resilience Station
32. Glass Terminal, Control Tower & Gate 1 Dreamliner
33. Evergreen Greenhouse Dome
34. Circular Recycling Center
35. Aurora Snow Stadium



DOWNTOWN

Nine Blocks, One Short Walk

Shops, schools, healthcare and homes within a short walk of each other, around the Aurora Roundabout.

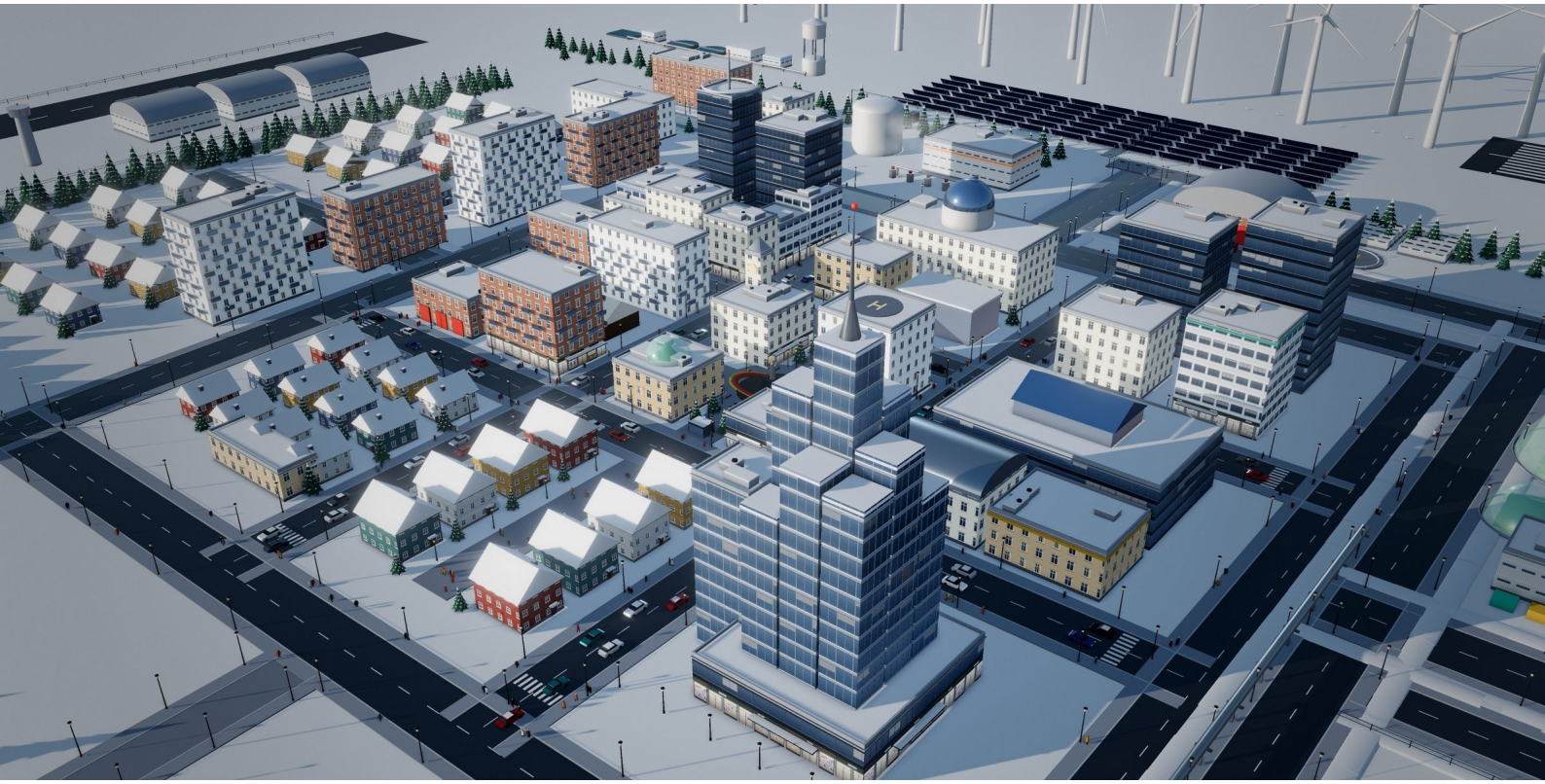
Downtown is laid out in nine blocks around the Aurora Roundabout and the President Ho Chi Minh Monument. The metro interchange lies under Liberty Mall, the tree-lined promenade through the four central blocks, which hosts festivals, markets and parades.



Liberty Mall at dusk, looking north to the President Ho Chi Minh Monument. Its granite plaza is warmed from below by the heating loop and never ices, and twin lanterns line the way.

DOWNTOWN

The Nine Blocks



Downtown from the south-east. The map below follows the city plan, with north at the top.

Polaris Quarter

Aurora Core – AI Control Tower · Aurora Haven Police Headquarters · Polaris Green Office Tower

Knowledge Quarter

Arctic University – Main Hall · Aurora Public Library

Financial District

Arctic Stock Exchange · Aurora Green Finance Hall · Snowline Courthouse · Aurora Business Hotel

West Market Street

West Market Fire & Ambulance Station · Aurora Night Market Hall · Polar Pharmacy

Civic Core & Liberty Mall

President Ho Chi Minh Monument · Civic Center · Snowflake General Hospital · Northern Lights School · Igloo Fresh · Aurora Cinema · Metro

East Gallery Avenue

Aurora Glacier Mall & Crystal Atrium · Snow Lantern Primary School · Northern Lights Music Hall

Snowpine Residences

Family homes · Pinecone Kindergarten · Snowpine Senior Care Home

Mount Aurora Approach

Mount Aurora Guesthouse · Southside Police & Mountain Rescue Post · spa shuttle street

Aurora 81 District

Aurora 81 – Polar Landmark Tower · Aurora 81 Residences (Twin Towers)

EVERYDAY LIFE

A Livable City Every Day



Snowpine Residences at dusk. Sidewalks warmed by district-heating return water stay snow-free, and spruces shelter the courtyards from the wind.

HOMES

Family houses in Snowpine Residences, the western eco-homes, towers such as the Aurora 81 Residences and Ice Crystal Apartments, co-living, student and senior housing.

LEARNING

Three kindergartens, Snow Lantern Primary School, the Northern Lights School for ages 6 to 18, and Arctic University.

HEALTH

Snowflake General Hospital with about 250 beds and a 24-hour emergency department, a campus clinic and a 24-hour pharmacy.

SHOPPING

Igloo Fresh Supermarket, Tundra Grocery, the night and fish markets and the Aurora Glacier Mall with its indoor atrium park.

CULTURE AND PARKS

Aurora Cinema, a music hall, a library open 24 hours in the polar night, the Snow Stadium and parks with heated benches.



The Mount Aurora Approach: low buildings keep the summit in view, and warm, low-glare lanterns do not hide the aurora.

250
hospital beds

900
pupils at Northern Lights School

5,000
students at Arctic University

8,000
seats in the Snow Stadium

1

SIGNATURE DISTRICT 1 · FUSION POWER STATION

The City's Own Sun

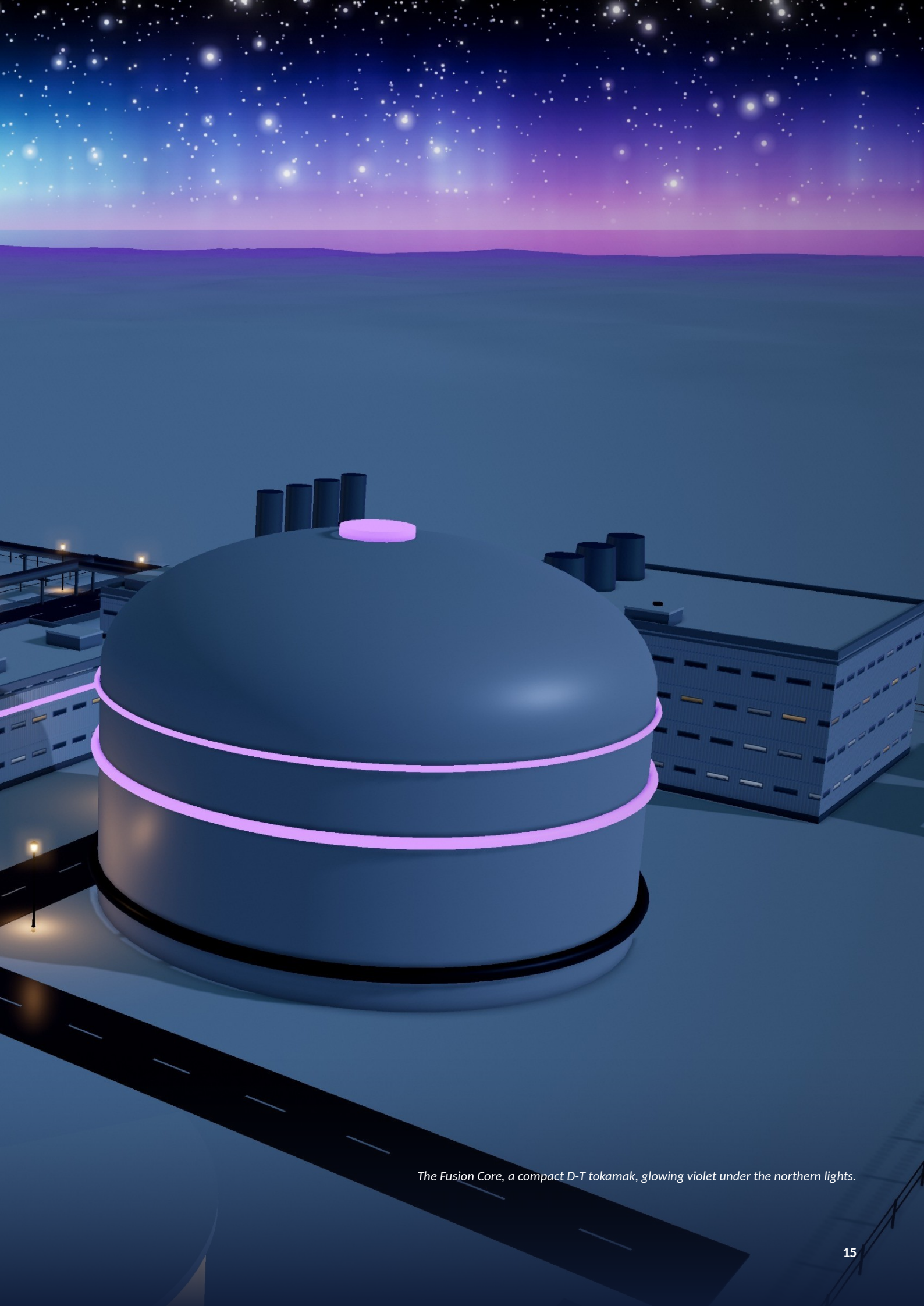
A compact tokamak gives Aurora Haven about 300 MW of electricity and 450 MW of district heat, steady through the polar night, with no CO_2 and no meltdown.



Our first signature district is the city's own sun. Stage 1 takes deuterium from snowmelt water, so the fuel literally falls from the sky as snow. Stage 2 breeds tritium in a lithium blanket, so none is imported.

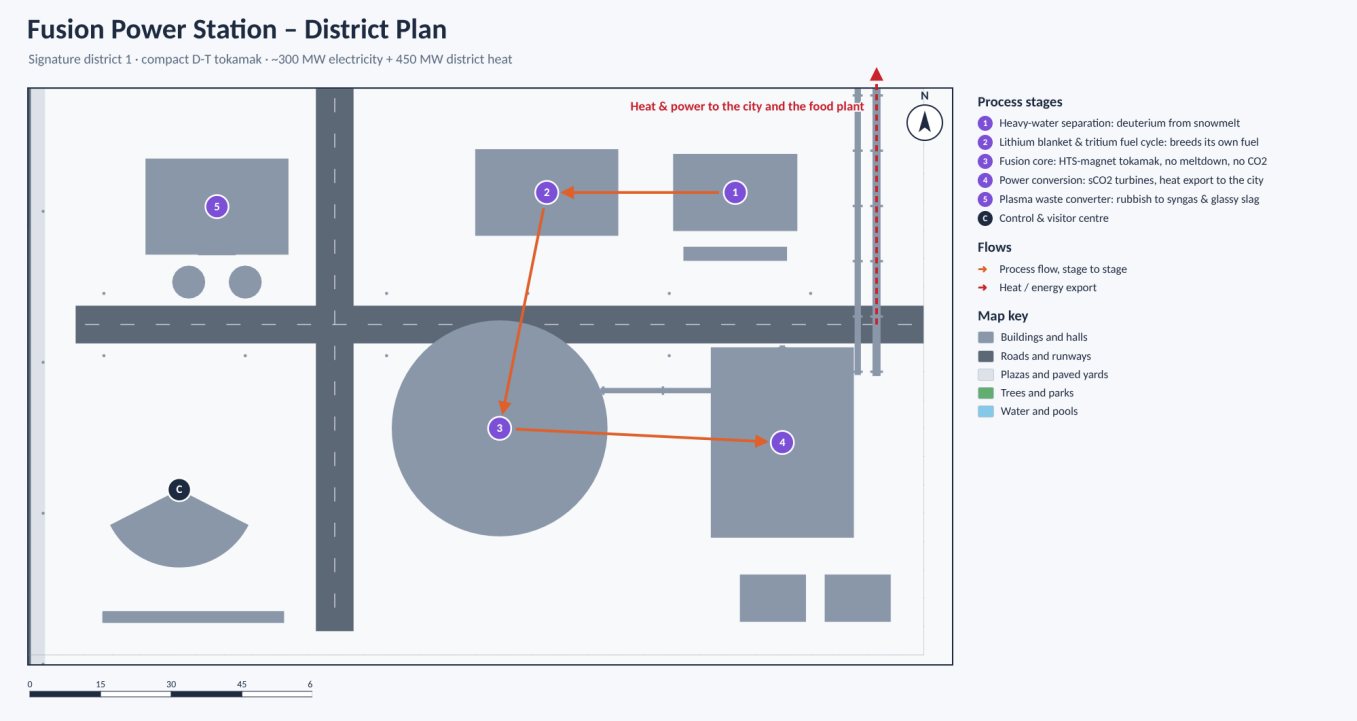
Stage 3 is the tokamak, where superconducting magnets hold a 150-million-degree plasma that makes about 800 megawatts of heat. Stage 4 turns that heat into about 300 megawatts of electricity and sends 450 megawatts into the district-heating loop.

The reactor runs on about 170 grams of fuel a day. If anything fails, the plasma simply goes out in a split second, so it cannot melt down or run away.



The Fusion Core, a compact D-T tokamak, glowing violet under the northern lights.

Five Stages, One Sun



Fusion Power Station – District Plan. Signature district 1: a compact D-T tokamak.

1 — Heavy-water separation

Takes deuterium from the deuterium-rich water left by the food plant's electrolysers and purifies it to 99.8% heavy water.

2 — Lithium blanket & tritium fuel cycle

Breeds slightly more tritium than the reactor burns, so none is imported. Recovered helium cools the hospital's MRI scanners.

3 — Fusion core: compact D-T tokamak

Superconducting magnets hold a 150-million-degree plasma. No CO_2, no long-lived waste, no meltdown.

4 — Power conversion & heat export

Supercritical-CO_2 turbines make about 300 MW of electricity and send about 450 MW of heat to the city.

5 — Plasma waste converter

Torches hotter than 5,000 °C turn about 20 t of rubbish a day into clean gas, metals and paving blocks.

C — Control & visitor centre

Operators and an AI digital twin can stop the plasma in milliseconds; a public gallery faces the reactor.

800 MW
of heat from the plasma

300 MW
of electricity

450 MW
of district heat for the city

170 g
of fuel a day, about 70 g of it deuterium

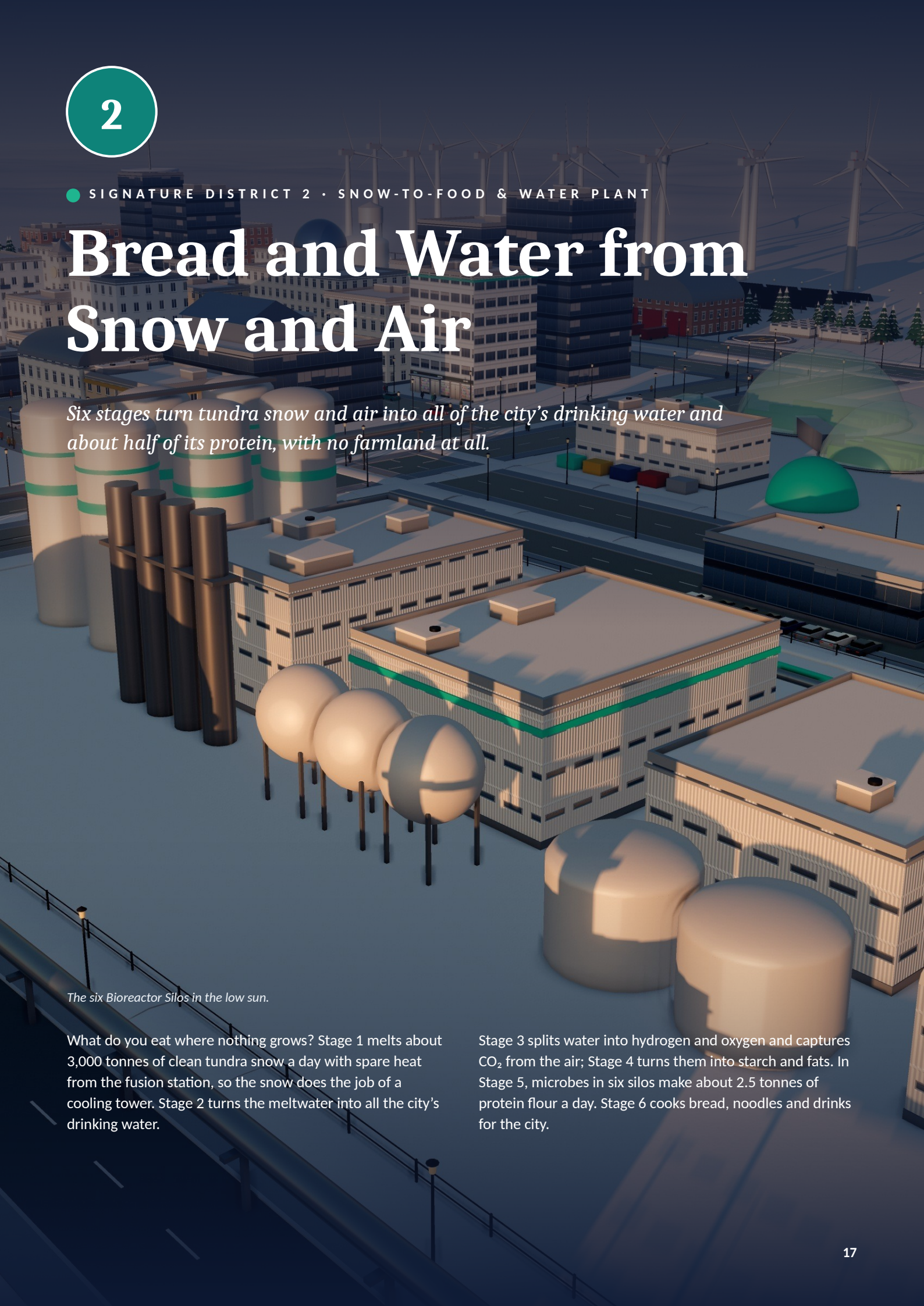
1 litre
of liquid helium a day for MRI scanners

2

SIGNATURE DISTRICT 2 · SNOW-TO-FOOD & WATER PLANT

Bread and Water from Snow and Air

Six stages turn tundra snow and air into all of the city's drinking water and about half of its protein, with no farmland at all.

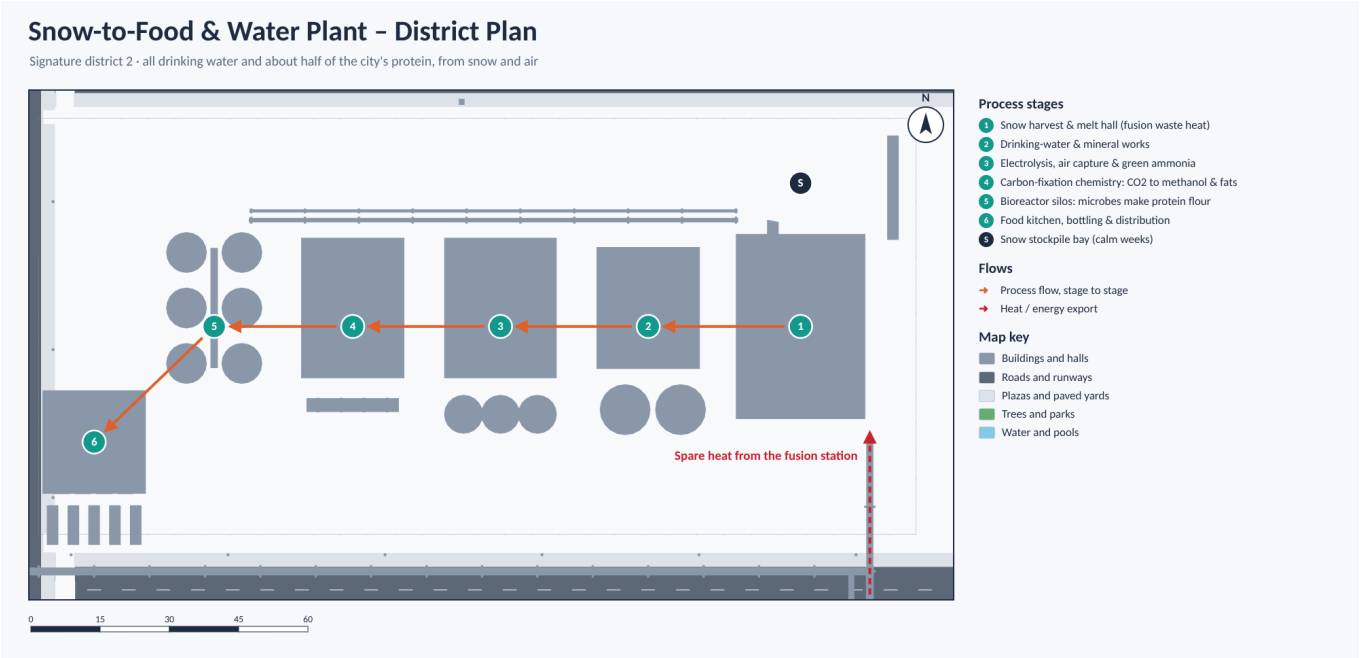


The six Bioreactor Silos in the low sun.

What do you eat where nothing grows? Stage 1 melts about 3,000 tonnes of clean tundra snow a day with spare heat from the fusion station, so the snow does the job of a cooling tower. Stage 2 turns the meltwater into all the city's drinking water.

Stage 3 splits water into hydrogen and oxygen and captures CO_2 from the air; Stage 4 turns them into starch and fats. In Stage 5, microbes in six silos make about 2.5 tonnes of protein flour a day. Stage 6 cooks bread, noodles and drinks for the city.

Six Stages from Snow to Supper



Snow-to-Food & Water Plant – District Plan. All drinking water and about half of the city's protein, from snow and air.

1 **Snow harvest & melt hall**

Electric harvesters gather wind-packed snow upwind of the city; about 14 MW of spare fusion heat melts it.

2 **Drinking-water & mineral works**

Membranes and UV light remove every germ, calcite adds calcium and magnesium, and sensors test the water every five minutes.

3 **Electrolysis, air capture & green ammonia**

About 5 t of hydrogen a day; the oxygen goes to the hospital and the waste converter.

4 **Carbon-fixation chemistry**

A copper catalyst makes methanol, and enzymes build it into starch about eight times faster than a maize field.

5 **Bioreactor silos**

Hydrogen-eating bacteria make protein flour; yeasts and algae make oils and vitamins, 24 hours a day.

6 **Food kitchen, bottling & distribution**

Bread, noodles, protein mince and dairy-style drinks, delivered overnight in heated electric vans.

3,000 t

of snow melted a day

2,500 m³

of drinking water a day, all of the city's tap water

20 t

of CO_2 captured from the air a day

2.5 t

of protein flour a day, as much as 600 hectares of soybeans

30 days

of meals and water for everyone in the Food Reserve Bunker

3

SIGNATURE DISTRICT 3 · MOUNT AURORA GEOTHERMAL SPA RESORT

A Warm Heart Inside the Ice

Inside a 35-metre snow peak shaped like Nepal's Ama Dablam, four geothermal spa levels stay at 28 °C all winter.



The four spa levels step out of the north face; the forecourt is warmed by the spa's used hot water, so guests walk ice-free from the shuttle to the door.

Guests enter through the North Portal, where a thermal revolving door and an air curtain keep the −45 °C air out. The peak's deep snow works as a blanket that holds in the geothermal heat, and sensors release the snow load safely before an avalanche can build.

Outside is the Aurora Lagoon, a 38-degree mineral pool for bathing under the northern lights. The spa fights the winter blues, brings tourists and income, and in an emergency can shelter up to 1,000 people.

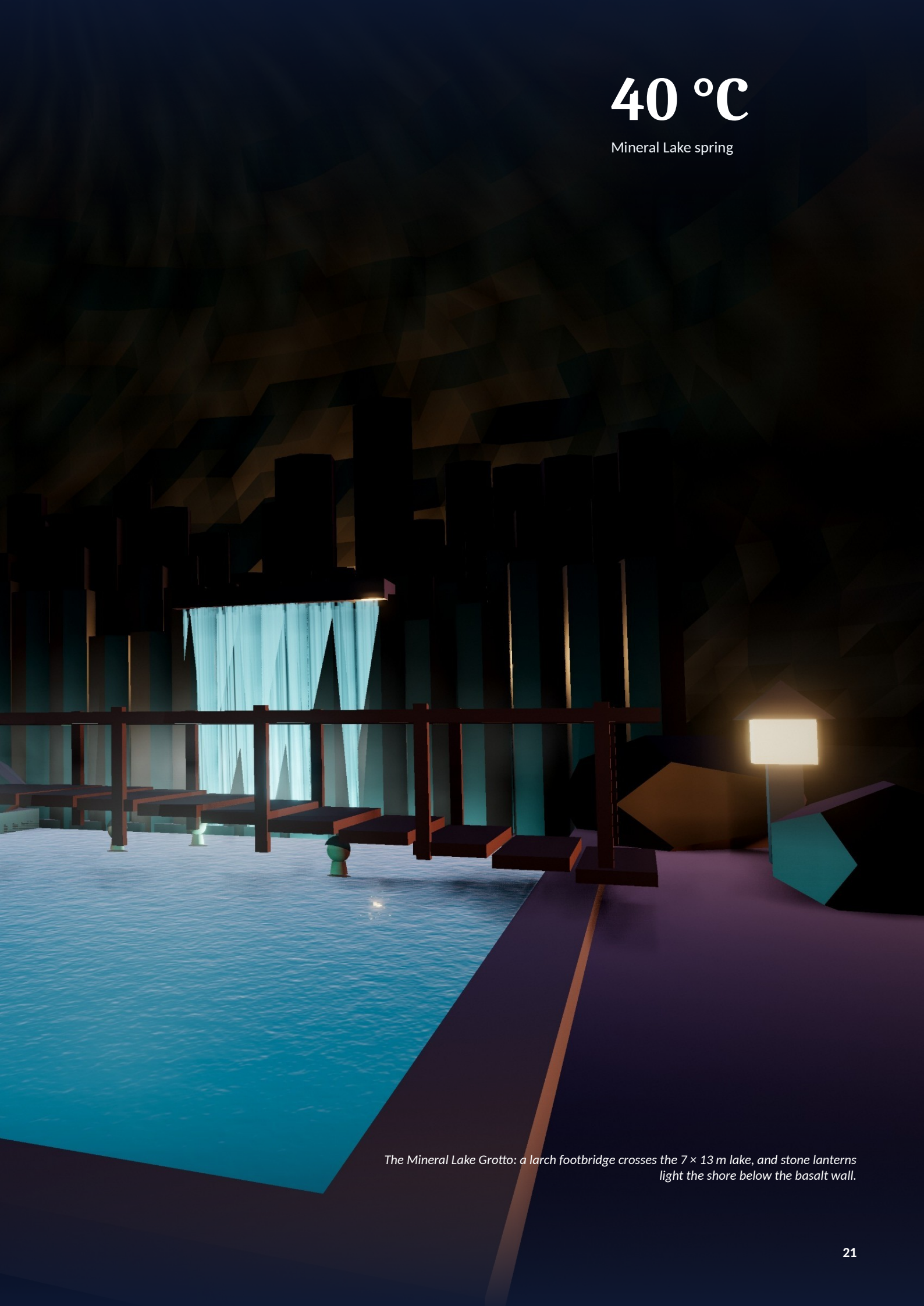
MOUNT AURORA · LEVEL 1

Inside the Mountain

The ground level and arrival floor of the spa: a welcome hall, two changing suites, a 7 × 13 m geothermal Mineral Lake grotto and an onsen garden with hot, warm and glacier-cold pools.

A 40 °C silica- and bicarbonate-rich spring rises from the geothermal field into the lake, spills over a basalt wall and flows on to the outdoor Aurora Lagoon.





40 °C

Mineral Lake spring

The Mineral Lake Grotto: a larch footbridge crosses the 7 × 13 m lake, and stone lanterns light the shore below the basalt wall.

Four Floors of Warmth



Mount Aurora Geothermal Spa Resort – Level Plans. The glass lift links all four levels.

LEVEL 1 · 0 M

Welcome Hall, Mineral Lake Grotto & Onsen Garden

A 40 °C silica-rich spring rises into a 7 × 13 m Mineral Lake; hot, warm and glacier-cold pools boost circulation.

LEVEL 2 · 4.5 M

Volcanic Mud Terraces

An 8 × 10 m mud field held at 39 °C. The mud is filtered and UV-sterilised every night and reused for months.

LEVEL 3 · 9 M

Treatment Suites & Himalayan Salt Sauna

Hot-stone massage, arctic kelp wraps and a 75 °C pink salt sauna heated by the geothermal loop.

LEVEL 4 · 13.5 M

Summit Sanctuary & Aurora Eye

A 35 °C float pool, a meditation circle and a heated glass bubble for watching the northern lights.

28 °C inside the spa

40 °C Mineral Lake spring

39 °C volcanic mud

75 °C salt sauna

35 °C float pool

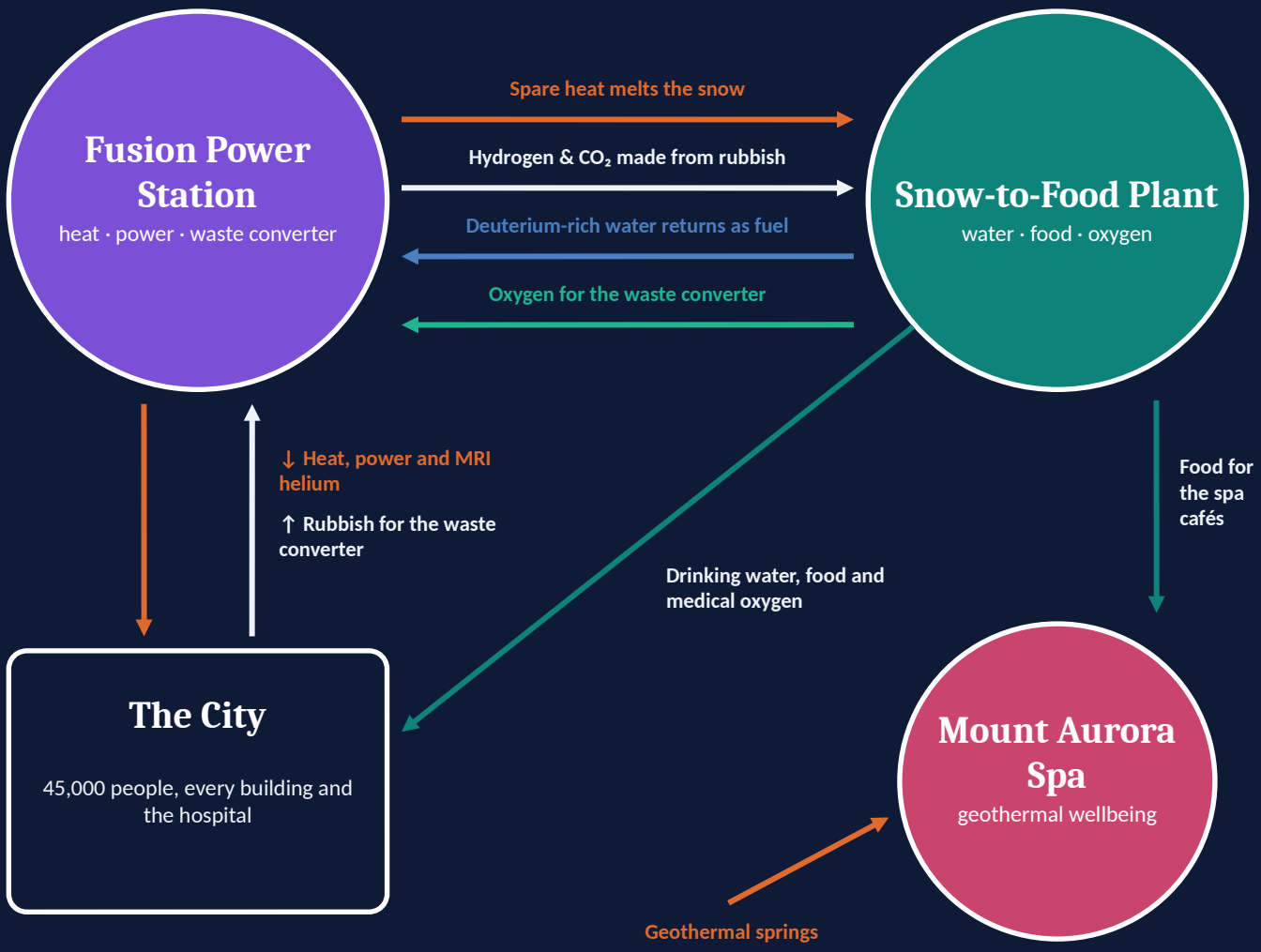
38 °C Aurora Lagoon

Route through the spa: North Portal → grotto → Welcome Hall → changing suites → Mineral Lake & Onsen Garden → glass lift → mud terraces → treatments & salt sauna → Summit Sanctuary and the Aurora Eye.

THE BIG IDEA

Three Districts, One Loop

The three signature districts do not work alone. Heat, water, oxygen and waste go round from one to the next instead of being thrown away.



Heat

Spare heat from the fusion station melts the snow at the food plant, so the snow does the job of a cooling tower.

Water

The food plant's electrolysers leave slightly deuterium-rich water, which goes back to the fusion station as fuel.

Oxygen

The same electrolysers supply oxygen to the plasma waste converter and the hospital.

Waste

The converter turns rubbish into hydrogen and CO_2 for the food lines, and its slag becomes paving.

Wellbeing

The spa turns geothermal heat into rest and health, and its restaurants serve food from the plant.

Energy for the Polar Night



Northern Renewable Park in the low sun: the Ice-Proof Wind Farm behind the Bifacial Solar Farm.

The sun disappears for two months, so the city uses sources that cover for each other.

Steady

Fusion

Steady power and heat all year, even in total darkness.

60 MW

Ice-proof wind farm

Heated blades keep turning in freezing fog; most power comes in stormy winter.

+25%

Bifacial solar farm

Snow reflects light onto the panel backs; 24-hour power in the midnight-sun months.

24/7

Geothermal plant

Clean electricity and hot water around the clock, and the second heat source.

200 MWh

Battery bank

Recycled car batteries store spare wind and solar power for calm, dark days.

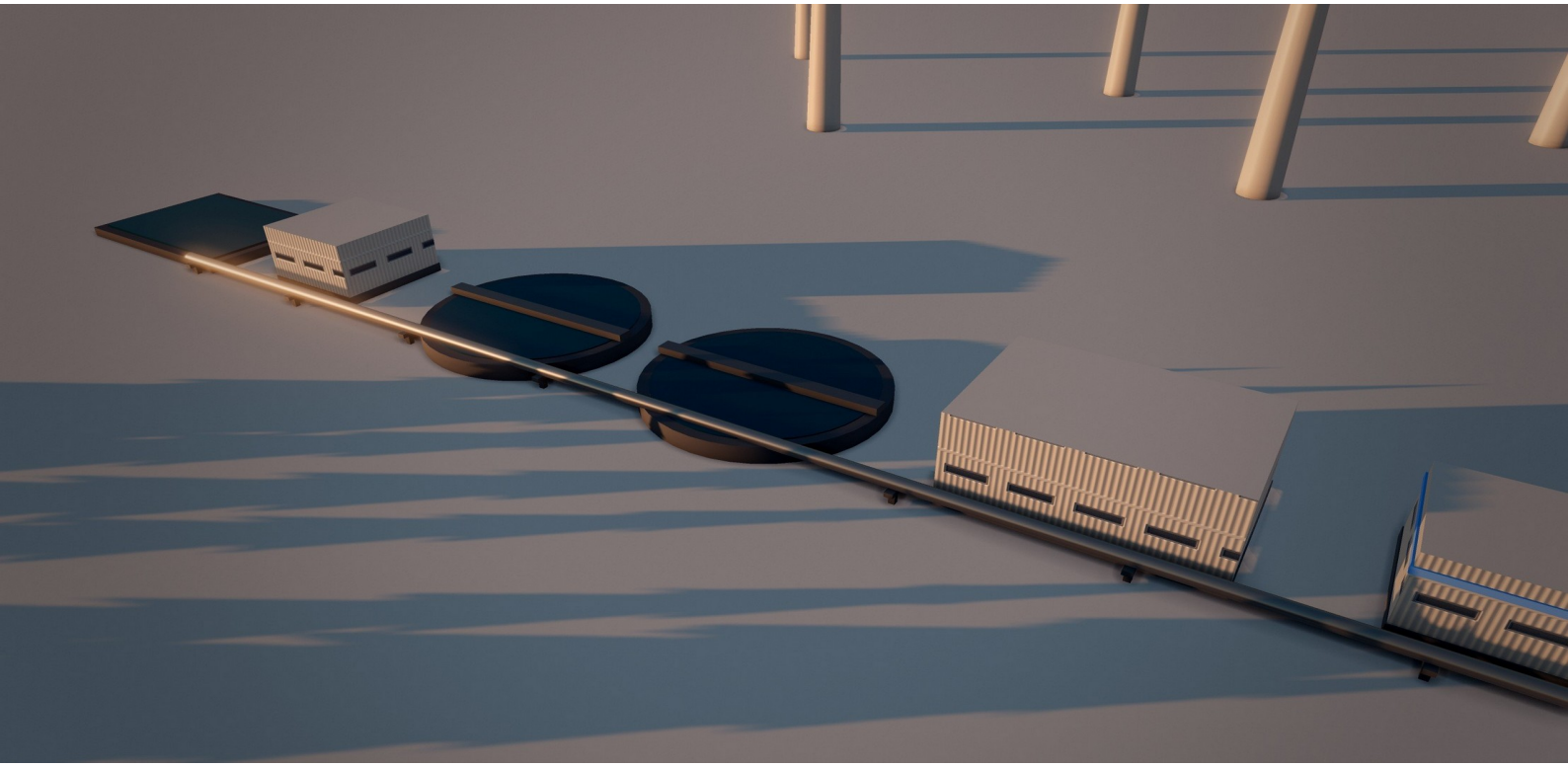
12 h

Heat storage tank

Keeps homes warm if a heat source stops, losing less than 1% of its heat a day.

WATER AND WASTE

Nothing Is Wasted



The six-stage Water Recycling Line in the Northern Renewable Park, with the wind farm behind.

THE SIX-STAGE WATER RECYCLING LINE

1. **Snowmelt & greywater intake**
2. **Intake pumps & bar screen**
3. **Settling basins**
4. **Sand & membrane filters**
5. **UV disinfection**
6. **Recycled water tower**

In Aurora Haven, water goes round and round. Clean snow becomes drinking water at the Snow-to-Food plant. Used greywater from homes flows to the Water Recycling Line: screens, settling basins, sand and membrane filters, UV light, and finally a tower that feeds toilets, laundries, greenhouses and fire hydrants.

Dirty street snow is melted separately and never reaches the taps.

The Circular Recycling Center sorts about 30 tonnes of materials a day with AI optical sorters and robot arms. Food scraps become biogas, and the Plasma Waste Converter handles what is left, turning it into clean gas and building blocks.

Nothing goes to landfill in the permafrost.

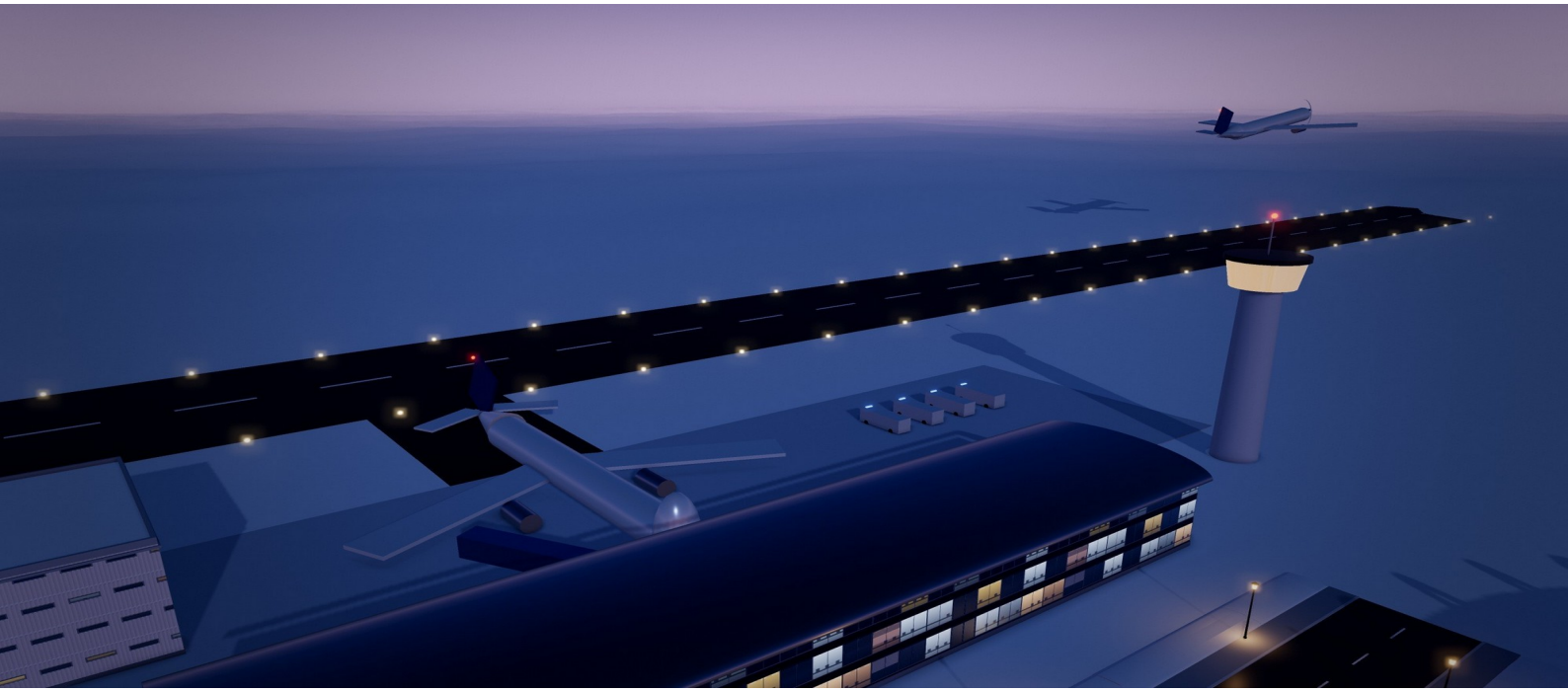
95%
of water recycled

30 t
of materials sorted a day

20 t
of rubbish a day turned into gas and blocks

0
landfill in the permafrost

Getting Around Fossil-Free

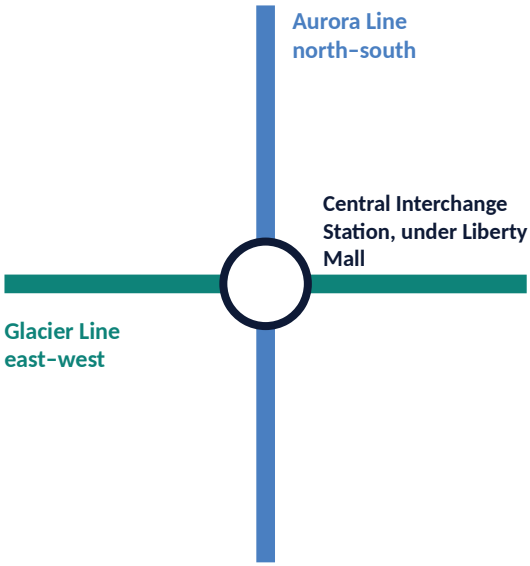


Compact Polar Airport at dusk: the glass terminal, the control tower and Flight VN-81 climbing out.

Transport in Aurora Haven is electric and built for the cold. The heart of it is the metro: the north–south Aurora Line and the deep east–west Glacier Line meet at the Central Interchange Station under Liberty Mall. The driverless trains run underground, so blizzards never stop them.

On the streets there are electric trams, buses and shared robotaxis, and people wait in heated bus shelters. The roads are warmed from below, so no salt is needed, and an outer ring keeps trucks out of downtown.

Electric cars charge in the heated bays of the EV & Hover-Car Charging Hub, and the airport, with its heated runway and e-fuel planes, keeps the city supplied through deep winter.



3 min
Aurora Line trains at rush hour

4 min
Glacier Line; every 90 s after big stadium events

30%
of train energy returned by braking

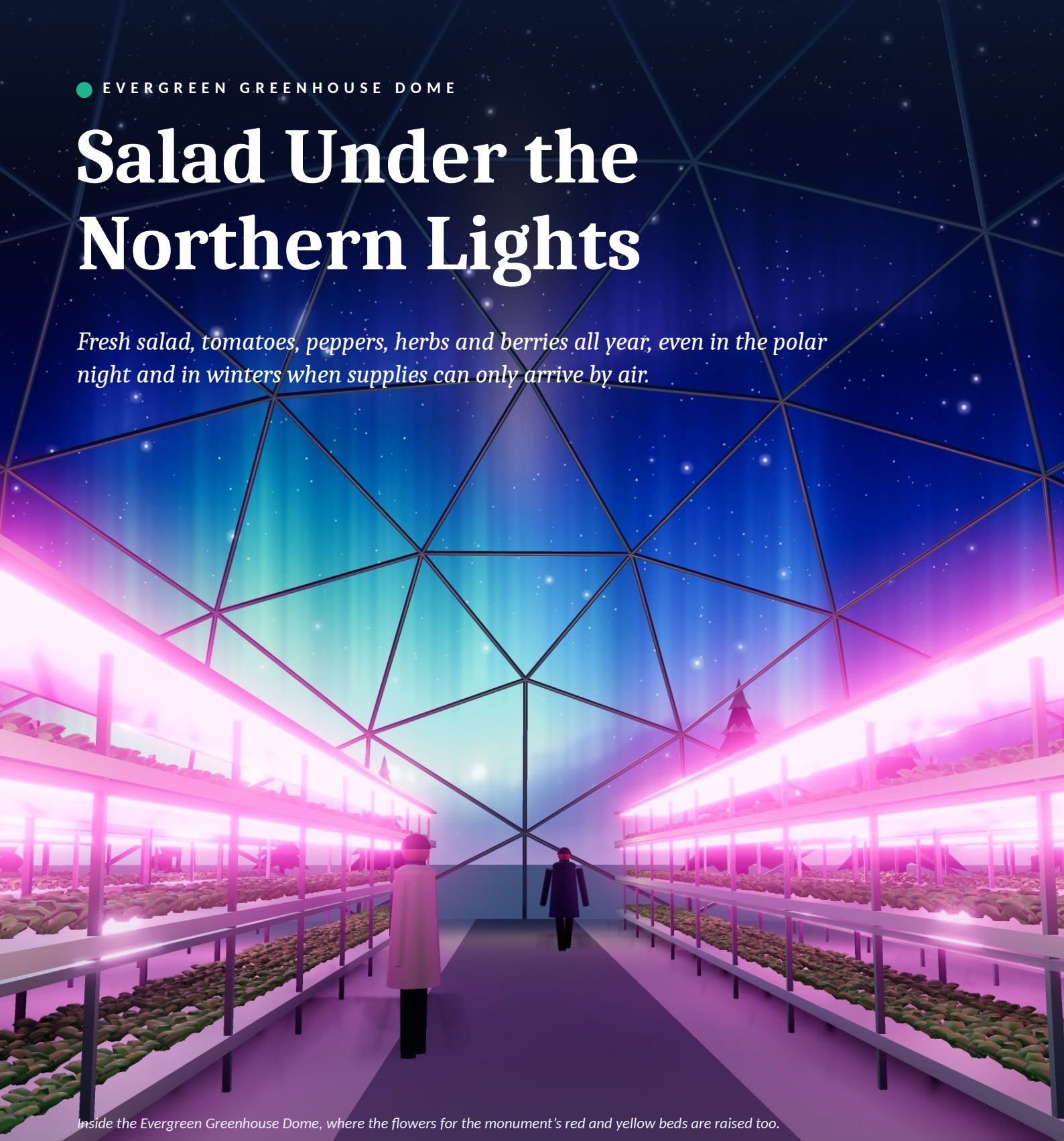
2,800 m
heated airport runway

20 km/h
downtown speed limit

EVERGREEN GREENHOUSE DOME

Salad Under the Northern Lights

Fresh salad, tomatoes, peppers, herbs and berries all year, even in the polar night and in winters when supplies can only arrive by air.



Inside the Evergreen Greenhouse Dome, where the flowers for the monument's red and yellow beds are raised too.

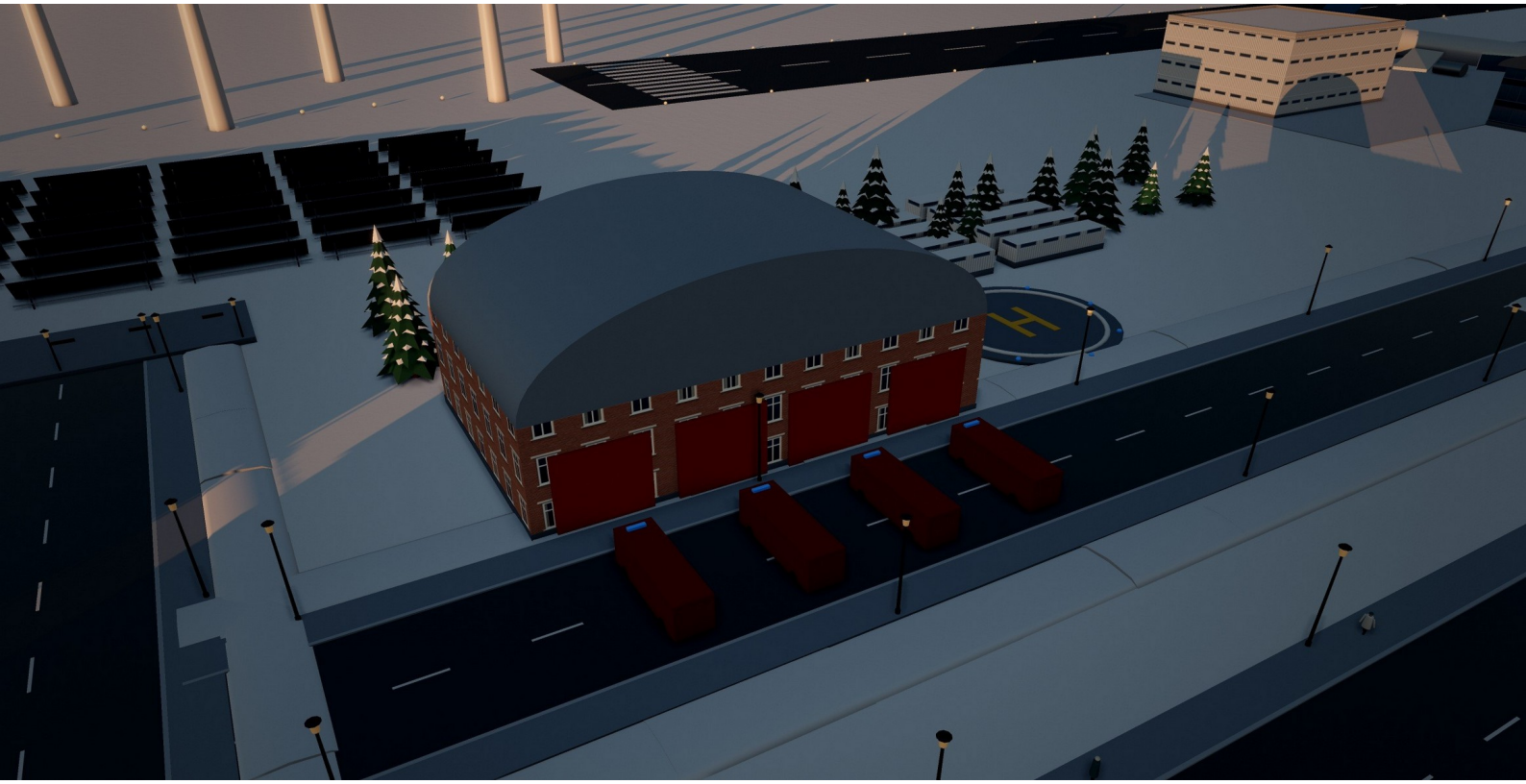
The dome is heated by the district-heating loop and wrapped in thermal curtains at night. LED grow lights give the plants 18-hour "days" on stacked hydroponic racks, and CO_2 from the biogas plant and the waste converter boosts growth.

18 h
LED "days" for the plants

90%
less water than fields

SAFETY

Safe and Ready for Any Storm



The Polar Rescue and Resilience Station in the Energy Hub, north of downtown, with its helipad.

In the polar night, safety comes first. The Polar Rescue and Resilience Station is home to the fire brigade, ambulances, ice rescue and the 112 emergency centre, and its vehicles leave within 60 seconds.

The West Market Fire & Ambulance Station reaches any downtown address in about four minutes, and a rescue helicopter is ready day and night.

Under Liberty Mall, four underground shelters can keep every resident safe for 30 days. Aurora Core, the AI control tower, forecasts blizzards up to 48 hours ahead and alerts every phone.

Cafés and lobbies act as warm points, and the air base runway is a backup for supply flights.

60 s
for rescue vehicles to leave

4 min
to any downtown address

10 min
for the rescue helicopter to be airborne

4
underground shelters

30 days
of supplies for every resident

48 h
blizzard forecasts

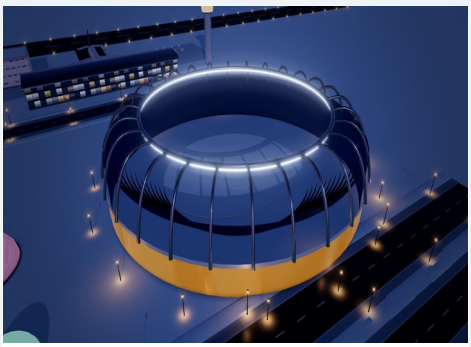
EXPLORE

Take the Tour: Eight Stops

A walk around Aurora Haven in eight stops, from the civic heart of downtown out to the plants, the snow peak and the windy northern edge.

1. **President Ho Chi Minh Monument**
   The civic heart of the city on the Aurora Roundabout.
2. **Metro Entrance – Central Interchange Station**
   Escalators down to the metro and the shelters.
3. **Snowflake General Hospital**
   About 250 beds and a 24-hour emergency department.
4. **Fusion Core – Compact D-T Tokamak**
   The city's own sun.
5. **Snow Harvest & Melt Hall**
   Melts about 3,000 t of snow a day with spare fusion heat.
6. **Mount Aurora – Snow Peak & Hanging Glacier**
   The 35 m peak with the spa inside.
7. **Ice-Proof Wind Farm**
   Two rows of turbines making about 60 MW.
8. **Polar Rescue and Resilience Station**
   Home of the 112 emergency centre.

If there is time: the Aurora Lagoon, the Bifacial Solar Farm and the Recycled Water Tower.



Aurora Snow Stadium at dusk.

BEHIND THE BUILD

35
key places, numbered on the master plan

9
downtown blocks around the Aurora Roundabout

1 m
per unit on every plan, drawn from the 3D model

MEETING THE BRIEF

1. **Smart design**
   polar answers to cold, darkness, wind and permafrost
2. **Sustainability**
   clean energy, closed loops, green spaces, electric transport
3. **Livability**
   homes, schools, hospital, shops, parks, cinema, roads
4. **Detail**
   one master plan, two district plans and four spa level plans

Places to Visit

The numbers match the markers on the city master plan on pages 8–9.

1 President Ho Chi Minh Monument
The city's civic heart on Aurora Roundabout: ceremonies, national holidays and walking tours.

2 Aurora Civic Center & Municipal Hall
Town hall for the city council, ID cards, permits, housing and family services.

3 Snowflake General Hospital
Main hospital with about 250 beds and a 24-hour emergency department.

4 Northern Lights School & Botanical Conservatory
Primary and secondary school for about 900 pupils aged 6–18.

5 Igloo Fresh Supermarket & Food Hall
Downtown's main supermarket and food court; about half its food is made in the city.

6 Aurora Cinema & Performing Arts Center
Cinema and theatre for films, concerts, ballet and school shows.

7 Underground Metro & Emergency Shelters
Metro interchange linked to four shelters that last 30 days.

8 Aurora Core – AI Control Tower
Operations centre where an AI digital twin runs traffic, heat, power and alerts.

9 Aurora Haven Police Headquarters
Central police station with a 24-hour police control room.

10 Arctic University – Main Hall
Lecture halls for about 5,000 students of polar engineering, fusion, food science, medicine and climate.

11 Aurora Public Library
Free library, open 24 hours in the polar night; lends skis, telescopes and light-therapy lamps.

12 West Market Fire & Ambulance Station
Downtown fire and ambulance branch that reaches any downtown address in about 4 minutes.

13 Aurora Glacier Mall & Crystal Atrium
Largest downtown mall, with an indoor atrium park of real trees.

14 Snow Lantern Primary School
Primary school for about 450 children aged 6–11 from the southern districts.

15 Pinecone Kindergarten
Kindergarten and day nursery for about 40 children aged 1–6.

16 Aurora 81 – Polar Landmark Tower
Tallest tower and skyline symbol: offices, hotel and sky apartments, net-zero energy.

17 Aurora Snow Garden Park
Public park with heated benches, a sledging slope and a snow-sculpture trail.

18 Mount Aurora – Snow Peak & Hanging Glacier
35 m landmark peak with the four-level spa inside.

19 Summit Sanctuary & Aurora Eye
Float pool, meditation circle and the Aurora Eye viewing bubble for the northern lights.

20 Aurora Lagoon – Outdoor Geothermal Mineral Pool
38 °C open-air silica pool for bathing under the northern lights.

21 Fusion Core – Compact D-T Tokamak
150-million-degree plasma giving about 800 MW of heat; it cannot melt down.

22 Power Conversion & Heat Export Hall
Makes about 300 MW of electricity and sends about 450 MW of heat to the city.

23 Plasma Waste Converter
Turns about 20 t of rubbish a day into clean gas, recovered metals and building blocks.

24 Snow Harvest & Melt Hall
Melts about 3,000 t of clean snow a day with spare fusion heat.

25 Drinking-Water & Mineral Works
Makes about 2,500 m^3 of drinking water a day, all of the city's tap water.

26 Bioreactor Silos
Six silos where microbes make about 2.5 t of protein flour a day.

27 Ice-Proof Wind Farm
Two rows of ice-proof turbines making about 60 MW.

28 Bifacial Solar Farm
Double-sided panels that make power 24 hours a day in the midnight-sun months.

29 Water Recycling Line – Recycled Water Tower
Stores recycled water for toilets, laundries, street cleaning, irrigation and hydrants.

30 Geothermal Power Plant
Clean electricity and hot water around the clock; the second heat source.

31 Polar Rescue and Resilience Station
Fire, ambulance and ice-rescue headquarters, home of the 112 emergency centre.

32 Glass Terminal, Control Tower & Gate 1 Dreamliner
Passenger and cargo airport; the city's supply link in deep winter.

33 Evergreen Greenhouse Dome
Grows fresh vegetables and berries all year under LED light.

34 Circular Recycling Center
Sorts, repairs and recycles about 30 t of materials a day.

35 Aurora Snow Stadium
About 8,000 seats for snow sports, school sports days and the Midwinter Aurora Festival.

KEY FIGURES

Aurora Haven by the Numbers

THE CITY	**45,000** residents	**9** downtown blocks	**35** key places
CLIMATE	**−45 °C** winter lows	**2 months** of polar night	**120 km/h** blizzards
FUSION POWER STATION	**300 MW** electricity	**450 MW** district heat	**99.8%** heavy-water purity
SNOW-TO-FOOD & WATER PLANT	**5 t** hydrogen a day	**4 t** starch a day	**1 t** fat a day
ENERGY AND STORAGE	**60 MW** wind farm	**200 MWh** battery bank	**<1%** heat lost a day by the heat tank
MOUNT AURORA SPA	**4** levels, 0 to 13.5 m	**28 °C** all winter	**1,000** people sheltered in an emergency
LIVABILITY	**600** Aurora 81 apartments	**450** pupils at Snow Lantern Primary	**3** kindergartens
TRANSPORT	**2** metro lines	**5 min** spa shuttle	**20 min** airport shuttle

All figures are design values for the Aurora Haven city project.

UNTIL NEXT TIME

Warm, Green and Ready

Aurora Haven does not just survive the polar environment; it thrives in it. Heat from fusion and geothermal energy keeps every building warm, and clean energy and closed loops mean no fossil fuel and almost no waste. Most importantly, it is a place where 45,000 people can live, learn, work and watch the northern lights together.

AURORA HAVEN

[Group name] · [Member names]
[Class] · [Date]
Modelled in 3D with three.js
Polar / Ice City

